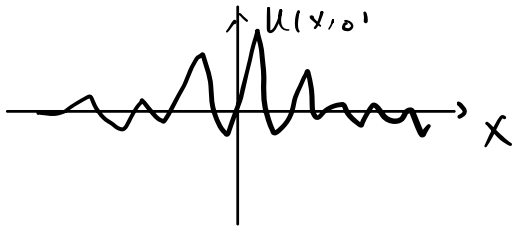
Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 09: Group velocity 2; Waveguide

Instructor: Zihan Wang

Example: Gaussian modulated oscillation

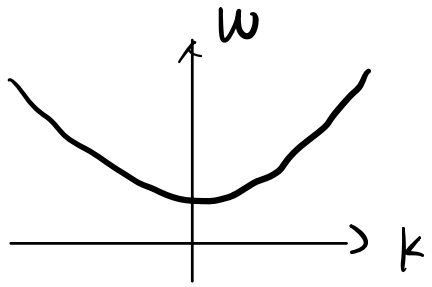
Initial conditions:



$$u(x, 0) = e^{-x^2/2L^2} \cos k_0 x \qquad \frac{\partial u}{\partial t}(x, 0) = 0$$

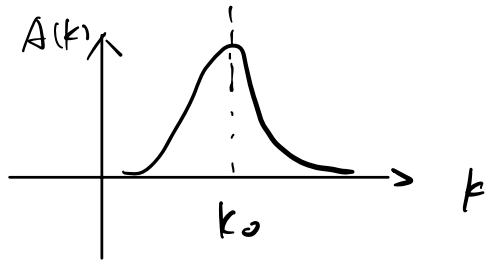
Dispersion relation:

$$\omega(k) = \nu\left(1 + \frac{a^2 k^2}{2}\right)$$



Group velocity:

$$v_g = \frac{dw}{dk} = \nu a^2 k$$



Example: Gaussian modulated oscillation

$$u(x, t) = \frac{1}{2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} A(k)e^{ikx - i\omega(k)t}\, dk + \text{c.c.}$$

$u(x,0)$

$$A(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ikx}\underline{e^{-x^2/2L^2}\cos k_0 x}\, dx$$

$$= \frac{L}{2}\left[e^{-(L^2/2)(k-k_0)^2} + e^{-(L^2/2)(k+k_0)^2}\right]$$

Example: Gaussian modulated oscillation

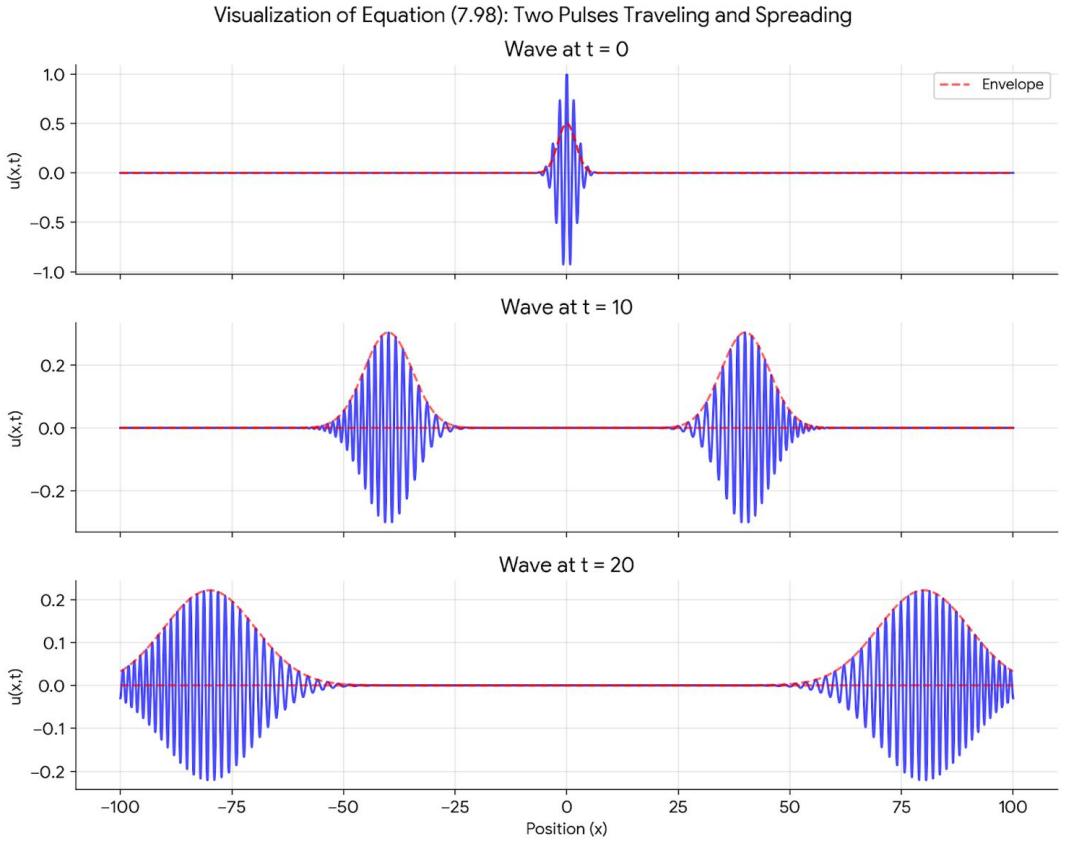
$$u(x,t) = \frac{L}{2\sqrt{2\pi}} \operatorname{Re} \int_{-\infty}^{\infty} \left[e^{-(L^2/2)(k-k_0)^2} + e^{-(L^2/2)(k+k_0)^2} \right] e^{ikx - i\nu t[1+(a^2k^2/2)]}\, dk$$

$$u(x,t) = \frac{1}{2}\operatorname{Re}\left\{ \frac{\exp\left[-\dfrac{(x - \nu a^2 k_0 t)^2}{2L^2\left(1 + \dfrac{ia^2\nu t}{L^2}\right)} \right]}{\left(1 + \dfrac{ia^2\nu t}{L^2}\right)^{1/2}} \exp\left[ik_0 x - i\nu\left(1 + \frac{a^2k_0^2}{2}\right)t \right] + (k_0 \to -k_0) \right\}$$

$v_g|_{k=k_0}$

$\omega(k_0)$

Example: Gaussian modulated oscillation



Spreading of a pulse in dispersive medium

- In terms of space

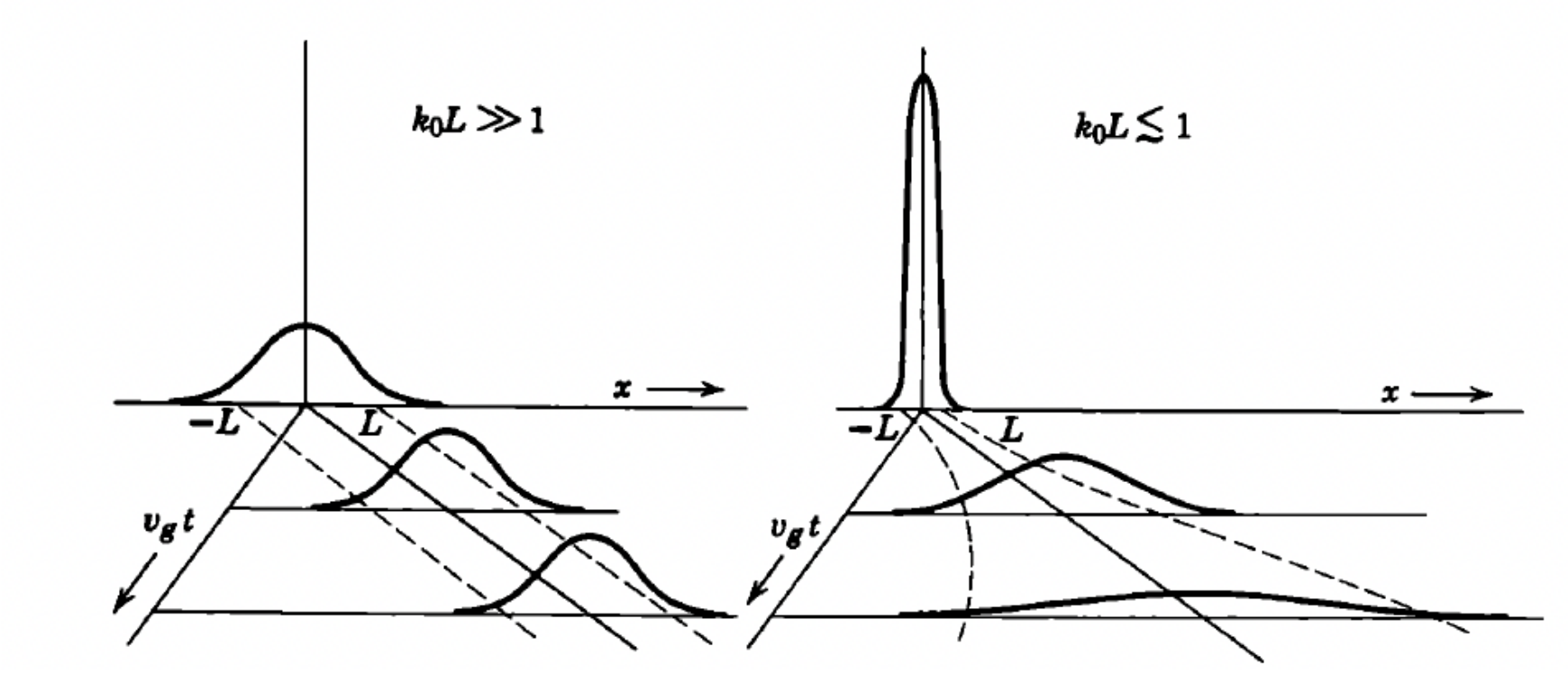
$$\text{Width} = \left(L^2 + \left(\frac{a^2 v t}{L} \right)^2 \right)^{1/2}$$

$$L \to 0, \quad \text{width} \approx \frac{a^2 v t}{L}$$

- In terms of time

$$L \to \infty, \quad \text{Width} = L$$

Spreading of a pulse in dispersive medium



Low frequencies EM waves transmit (wavelengths longer than meters)

- Line of sight: This refers to radio waves that travel in a straight line from the transmitting antenna to the receiving antenna. On the surface of the Earth, line of sight propagation is limited by the **visual horizon** to about 64 km (40 mi).
- Ground waves: At lower frequencies below 2 MHz, in the medium wave and longwave bands, due to diffraction vertically polarized radio waves can bend over hills and mountains, and propagate beyond the horizon, traveling as surface waves which follow the contour of the Earth.
- Skywaves: At medium wave and shortwave wavelengths, radio waves reflect off conductive layers of charged particles (ions) in a part of the atmosphere called the ionosphere. So radio waves directed at an angle into the sky can return to Earth beyond the horizon; this is called "skip" or "skywave" propagation.

High frequencies EM waves transmit

- At high frequencies (wavelengths shorter than meters), atmospheric gases begin absorbing waves, so the range of communication systems decreases with increasing frequency. Above 300 GHz, in the terahertz band, virtually all the power is absorbed within a few meters, so the atmosphere is effectively opaque.
- Thus, we need waveguide.

Cylindrical Cavities and Waveguides

- Propagation or excitation of electromagnetic waves in hollow metallic cylinders.
- If the cylinder has end surfaces,
- it is called a cavity;
- otherwise, a waveguide.

Propagation of EM waves in hollow metallic cylinders

- Perfect conductors.
- Cross-sectional size and shape are assumed constant along the cylinder axis.



Governing Equations for sinusoidal time dependence

$\vec{E}, \vec{B} \propto e^{-i\omega t}$

No source in the wave guide.

$$\begin{cases} \nabla \times \vec{E} = i\omega \vec{B} \\ \nabla \times \vec{B} = -i\mu\varepsilon\omega \vec{E} \\ \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \end{cases}$$

$$\begin{cases} \vec{E}_{\parallel} = 0 \\ \vec{B}_{\perp} = 0 \end{cases}$$

On S, what is the $\vec{E}, \vec{B}$?

Boundary conditions for waveguide

- Two ways to think about this
- 1. Reflection and transmission r ?
- 2. EM fields in the perfect conductor + Maxwell Equations

s pol : $r = -1$, $\vec{E}_c = 0$, $\vec{B}_c = 0$ in conductor

p-pol : $r = 1$.

Boundary conditions for waveguide

Method 2:

$$\begin{cases} \nabla \cdot \vec{D} = \dfrac{\rho}{\varepsilon_0} \\ \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{B} = \mu_0 \vec{j} \end{cases} \Rightarrow \begin{cases} \vec{n} \cdot (\vec{D} - \vec{D}_c) = \dfrac{\Sigma}{\varepsilon_0} \\ \vec{n} \times (\vec{E} - \vec{E}_c) = 0 \\ \vec{n} \cdot (\vec{B} - \vec{B}_c) = 0 \\ \vec{n} \times (\vec{B} - \vec{B}_c) = \mu_0 \vec{k} \end{cases}$$

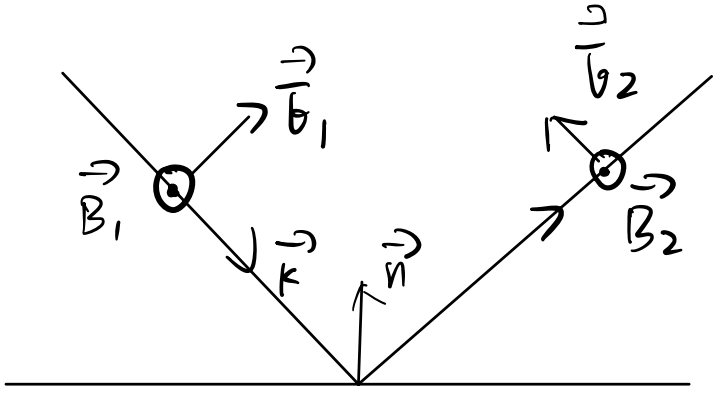
$$\Rightarrow \quad \vec{B}_\perp = 0 , \quad \vec{B}_{//} \neq 0 ,$$

$$\vec{E}_\perp \neq 0 , \quad \vec{E}_{//} = 0$$

Boundary conditions for waveguide

Method 1 : p-pol , r=1 .

$\vec{B} = \vec{B}_1 + \vec{B}_2 \qquad \vec{b} = \vec{b}_1 + \vec{b}_2$



$$\begin{cases} (\vec{B}_1 + \vec{B}_2) \times \vec{n} \neq 0 \\ (\vec{b}_1 + \vec{b}_2) \times \vec{n} = 0 \\ (\vec{b}_1 + \vec{b}_2) \cdot \vec{n} \neq 0 \end{cases}$$

$$\Rightarrow \vec{B}_{//} \neq 0 \cdot \vec{b}_{\perp} \neq 0, \ \vec{b}_{//} = 0$$

How to solve the Maxwell Eq?

- Let's start with the parallel component Ez and Bz.
- Let's use separation of variables

$$\begin{cases} \nabla \times \vec{E} = i\omega \vec{B} \\ \nabla \times \vec{B} = -i\mu\varepsilon\omega \vec{E} \\ \nabla \cdot \vec{E} = 0 \\ \nabla \cdot \vec{B} = 0 \end{cases} \Rightarrow \begin{cases} (\nabla^2 + \mu\varepsilon\omega^2)\, \vec{E} = 0 \\ (\nabla^2 + \mu\varepsilon\omega^2)\, \vec{B} = 0 \end{cases}$$

$$\begin{cases} \vec{E}_{\parallel} = 0 \\ \vec{B}_{\perp} = 0 \end{cases} \Rightarrow \begin{cases} (\nabla^2 + \mu\varepsilon\omega^2)\, E_z = 0 \\ (\nabla^2 + \mu\varepsilon\omega^2)\, B_z = 0 \end{cases}$$

2D Laplace Eq

$$E_z(\vec{x},t) = X(x)\,Y(y)\,Z(z)\,e^{-i\omega t}$$

$$\frac{1}{Z}\frac{d^2Z}{dz^2} = -k^2$$

$$Z = e^{ikz}$$

$$E_z(\vec{x},t) = XY\,e^{i(kz-\omega t)}$$

$$\nabla_t^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

$$\nabla^2 = \nabla_t^2 - k^2$$

$$\begin{cases} (\nabla_t^2 - k^2 + \mu\varepsilon\omega^2)\,XY = 0 \\ \vec{E}_{//} = 0 \Rightarrow E_z = 0 \text{ on } S \Rightarrow XY = 0 \text{ on } S \end{cases}$$